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Palaeomagnetism of the Tudor gabbro, Ontario: evidence for divergence between Grenvillia and interior Laurentia

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Summary. After thermal and alternating field (AF) cleaning, the characteristic high blocking temperature *A* component of natural remanent magnetization (NRM) of the Tudor gabbro of southern Ontario has a mean direction $D = 326°$, $I = -46°$ ($k = 132$, $\alpha_{95} = 4.8°$, $N = 8$ sites). The corresponding palaeopole, 133°E, 12°N ($dp = 4°$, $dm = 6°$), confirms the palaeopole 137°E, 17°N ($\alpha_{95} = 8.4°$) reported earlier by Palmer & Carmichael, based on AF cleaning only. The *A* NRM has unblocking temperatures >515–525°C which exceed the estimated 500°C peak temperature reached locally during ≈ 1050 Ma Grenvillian regional metamorphism. The *A* NRM therefore predates metamorphism and is probably a primary thermoremanence (TRM). The age of the Tudor NRM has previously been taken to be about 675 Ma, but recent $^{40}Ar/^{39}Ar$ dating by Baksi has shown that this is the time of post-metamorphic cooling to 200–250°C. Hornblendes record initial cooling of the intrusion to 590 ± 20°C at 1110 Ma and this is the best estimate of the age of the *A* remanence. Successful Thellier-type palaeointensity determinations on 11 Tudor samples confirm that the *A* NRM is a TRM and indicate a palaeofield at this time of 18–27 μT, about 50–70 per cent of the present field intensity at 27° magnetic latitude. The anomalous Tudor *A* palaeopole, which lies well to the west of both 1000–800 Ma Grenvillian palaeopoles and 1100–1050 Ma poles from Interior Laurentia, is interpreted as recording divergence between Grenvillia and Interior Laurentia just before the Grenvillian orogeny, rather than a post-metamorphic extension of the apparent polar wander path as previously assumed.

Key words: continental collisions, Grenville Province, Grenvillian orogeny, palaeofield intensity, palaeomagnetism, Precambrian

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Introduction

Penetrating the Grenvillian 'metamorphic veil' (Berger & York 1981) to recover the pre-orogenic palaeomagnetic and drift history of Grenvillia has proved unexpectedly difficult (Dunlop *et al.* 1980; Park & Emslie 1983). Numerous palaeomagnetic poles that were at first believed to 'see through' the Grenvillian orogeny almost certainly date from post-metamorphic regional uplift and cooling (e.g. Irving, Emslie & Ueno 1974; Buchan & Dunlop 1976; Symons 1978). They fall on the Grenville Loop of the Laurentian apparent polar wander path (APWP) in the approximate time interval 1000–800 Ma (Berger, York & Dunlop 1979; Dallmeyer & Sutter 1980). A recent example is the *A* palaeopole of the early Palaeohelikian (≈1700 Ma, U/Pb zircon) French River anorthosite (Stupavsky & Symons 1982), which falls on the Grenville Loop around 950 Ma.

In parts of the Hastings Lowlands of southern Ontario (Fig. 1), the metamorphic grade is unusually low for the Grenville Province and pre-metamorphic $^{40}Ar/^{39}Ar$ hornblende ages have been recovered. The Thanet gabbro has hornblende ages as old as 1200 Ma (Berger & York 1981), the Cordova gabbro has ages as old as 1150 Ma (Lopez-Martinez & York 1983) and hornblendes from the Tudor gabbro define an 1110 Ma isochron (Baksi 1982). These ages are essentially identical to the U/Pb zircon emplacement ages of the plutons. However, primary magnetizations have apparently not survived for the Thanet and Cordova bodies. Their high and medium blocking temperature natural remanent magnetizations (NRMs) give palaeopoles that fall on the usual Grenville path (Buchan 1978; Dunlop & Stirling 1985) and are therefore younger than 1000 Ma.

The Tudor gabbro, on the other hand, has long been known to have an anomalous palaeo-

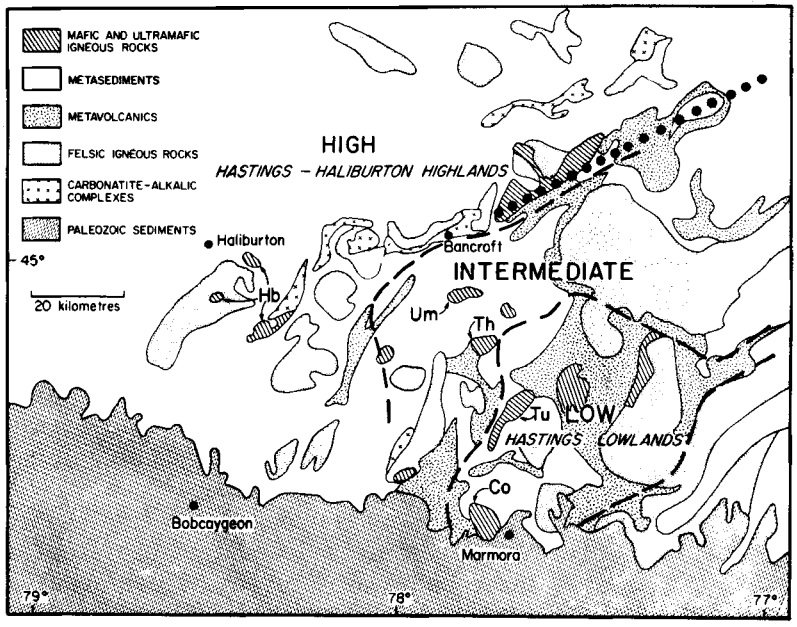


Figure 1. Geology and metamorphic zones in the Hastings–Haliburton area of southern Ontario, after Lumbers (1964) and Buchan (1978). The dashed lines are isograds separating regions of low (greenschist), intermediate (lower amphibolite) and high (upper amphibolite to granulite) metamorphic grade. Intrusions of palaeomagnetic interest referred to in this paper are Hb (Haliburton intrusions: Buchan & Dunlop 1976), Um (Umfraville gabbro; Symons 1978; Palmer *et al.* 1979), Th (Thanet gabbro; Buchan 1978); Tu (Tudor gabbro; Palmer & Carmichael 1973; this paper), and Co (Cordova gabbro; Dunlop & Stirling 1985).

pole (Hood 1958; Palmer & Carmichael 1973) that falls to the north and west of other Grenville poles. Hayatsu & Palmer (1975) determined a K/Ar whole-rock isochron age of 670 Ma for the Tudor body, which Baksi (1982) has shown dates the time of post-metamorphic cooling of plagioclases to about 200°C. This age has come to be associated with the palaeopole as well, and most palaeomagnetists have incorporated the Tudor pole on a post-Grenvillian extension of the APWP (e.g. Irving 1979; Roy & Robertson 1979; Buchan *et al.* 1983).

There are two problems with this traditional interpretation. If the Tudor NRM was frozen in thermally at temperatures around 200°C, it could hardly have escaped viscous overprinting in later geomagnetic fields (e.g. Pullaiah *et al.* 1975). If the NRM was acquired thermochemically, producing laboratory unblocking temperatures considerably higher than 200°C, the marked difference between the Tudor pole and late metamorphic *B* poles of the neighbouring Cordova, Thanet and Haliburton intrusions (Fig. 1) is hard to account for. All four bodies must have had parallel uplift and cooling histories. Apart from their proximity, $^{40}Ar/^{39}Ar$ step-heating analyses of plagioclases indicate cooling in all cases to $\approx 200°C$ around 700–800 Ma (Berger *et al.* 1979; Berger & York 1981; Baksi 1982; Lopez-Martinez & York 1983).

The other possibility is that the Tudor pole is unique because it alone is pre-metamorphic. This possibility is worth considering, since the Tudor lies in the lowest metamorphic zone in the Hastings Basin (Fig. 1) and was not reheated to as high temperatures as the Thanet and Haliburton stocks. The Cordova gabbro, which was reheated to temperatures similar to those experienced by the Tudor, has unmistakably suffered post-metamorphic thermochemical resetting of its entire NRM (Dunlop & Stirling 1985). A pre-metamorphic origin for part of the Tudor NRM cannot be taken seriously, therefore, unless this NRM fraction can be shown to be a thermoremanent magnetization (TRM) with blocking temperatures $\geqslant 500°C$, the peak metamorphic temperature reached locally.

Palmer & Carmichael (1973) unfortunately did not thermally demagnetize their samples. We therefore undertook a palaeomagnetic restudy of the Tudor gabbro and its contact rocks, concentrating on detailed thermal analyses of palaeofield directions and intensities.

Geological setting and sampling scheme

The regional geology of the Tudor gabbro (Fig. 2) has been described in detail by Lumbers (1967, 1969) and only a summary will be given here. The Grenvillian orogeny in the Hastings area had at least three phases. The first was deformation, largely monoclinal tilting, of thick volcanic and sedimentary sequences, of which the Tudor metavolcanics and Hastings marble are examples, between 1300 and 1250 Ma. This was followed by intrusion of post-kinematic granitic stocks like the Glanmire complex around 1250 Ma (Silver & Lumbers 1966) and of trondjemitic and mafic stocks like the Tudor gabbro around 1100–1200 Ma (S. B. Lumbers, cited in Symons 1978; Berger & York 1981; Baksi 1982). Finally erosion, uplift and slow cooling, either continuous or discontinuous, occurred from perhaps 1050–800 Ma or later.

Lumbers (1969) believes that burial and regional metamorphism coincided with the kinematic phase and preceded intrusion of all granitic and mafic stocks. Baer (1981) favours two separate thermal pulses. Whatever the timing of events, the area surrounding the Tudor gabbro was evidently never buried very deeply compared to other parts of the Grenville Province. The Tudor stock lies below both the oligoclase and zoned albite isograds (Lumbers 1969, chart A) and cannot have experienced metamorphic temperatures greater than 500°C at most. A reheating this mild would not necessarily have destroyed the contact meta-

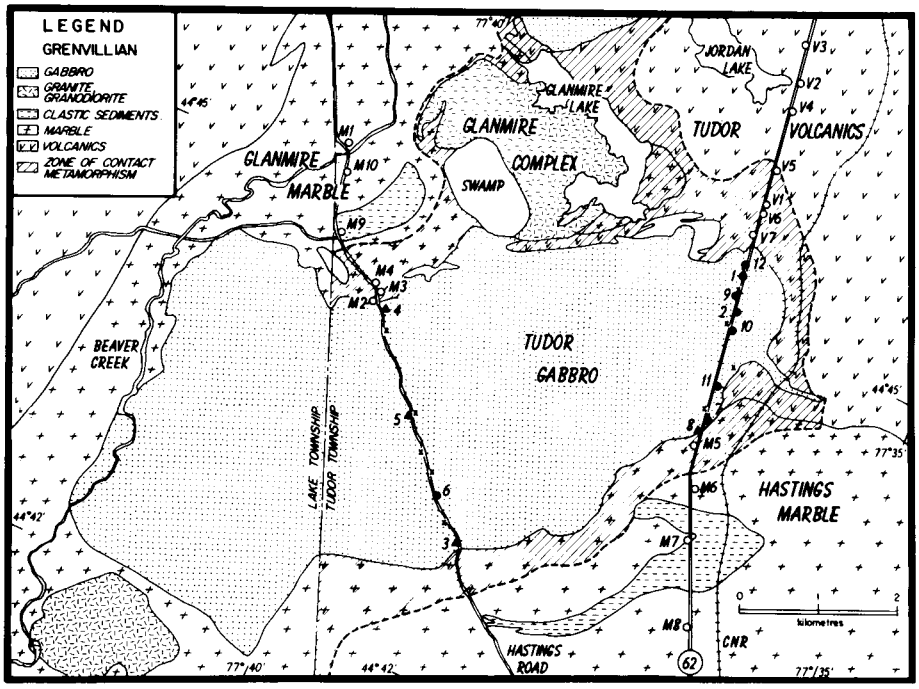


Figure 2. Generalized geological map of the Tudor gabbro and its host rocks, indicating the mapped limits of contact metamorphism around the stock. Sampling sites in the Tudor gabbro are labelled 1–12; circular and triangular symbols indicate sites that gave stable and unstable palaeomagnetic results respectively (most of the unstable sites are within 100 m of the margin of the intrusion); sites sampled by Palmer & Carmichael (1973) are shown by crosses. Sites in baked and unbaked country rocks are labelled V1–V9 for the Tudor volcanics (sites V8 and V9 are just off the upper edge of the map) and M1–M10 for the Glanmire and Hastings marbles.

morphic aureole around the Tudor body and emplacement of the body after the peak of regional metamorphism (Lumbers 1969) is not necessarily implied.

The main reason for favouring a pre-metamorphic age of intrusion is the following. Baksi (1982) finds that hornblendes in the Tudor gabbro cooled through 590°C around 1110 Ma. This temperature is about equal to the maximum blocking temperatures of the characteristic NRM observed in the present study. If the NRM is primary, its age must be approximately 1110 Ma, since the Tudor body was never subsequently reheated above about 500°C. We observe that the NRM is very consistently overprinted or remagnetized in the blocking temperature range below about 500°C. This overprinting we attribute to regional metamorphic heating.

This scenario, if it can be substantiated, is important because it would date the peak of Grenvillian metamorphism (in this area at least) more precisely than has been possible previously. The best proof would be to demonstrate a primary NRM by means of a positive baked contact test with the country rocks intruded by the Tudor. Failing this, the next best proof would be a demonstration that the characteristic NRM has the properties of a TRM, either primary or secondary, so that laboratory blocking temperatures are a reliable guide to peak temperatures experienced in nature.

Sixty-eight block samples were collected from 12 sites in the Tudor gabbro (Fig. 2). Sites were concentrated in the eastern lobe of the stock, which has a marked negative aeromagnetic anomaly, in fresh roadcuts along Hwy. 62. As in Palmer & Carmichael's (1973) study,

our best results came from these sites; samples taken along the former Hastings Settlement Road tended to have unstable behaviour.

We also made three traverses through the contact metamorphic aureole into unbaked or only slightly baked country rocks. Thirty-two samples of the Glanmire and Hastings marbles were collected at sites M1–M10, and 55 samples of the Tudor volcanics from sites V1–V9 were studied. Samples from seven of the volcanic sites were collected in an earlier field season by K. L. Buchan, C. J. Hale and J. M. Stirling.

Analytical procedures

Three or four 2.5 cm cylindrical cores were drilled in the laboratory from each block sample. NRMs in the course of stepwise alternating field (AF) and thermal demagnetization were measured on Schonstedt SSM-1 and modified Digico spinner megnetometers, both computer-interfaced. Except in cases where the NRMs were below instrumental noise levels ($< 10^{-4}$ A m^{-1}) or behaved erratically in the course of cleaning, every specimen was subjected to a full programme of thermal or AF demagnetization. The usual programme was 12 steps from 0 to 100 mT using a Schonstedt GSD-1 demagnetizer for AF cleaning and 16 steps from 20 to 640°C using a Schonstedt TSD-1 furnace for thermal cleaning.

Twenty-four Tudor gabbro samples had predominantly univectorial NRMs throughout thermal cleaning. Fresh specimens of these samples were used for Thellier-type determinations of palaeofield intensity. The procedure was to heat a specimen twice to the same temperature, cooling the first time in zero field and the second time in a field of known direction and intensity, in this case 50 μT. The purpose of the first heating is to demagnetize the NRM in steps, while the second heating allows one to deduce the partial TRMs acquired over the same temperature steps. The NRM lost and the partial TRM gained in a given temperature interval are proportional to each other only if the NRM itself originated as a TRM. If this turns out to be the case, the slope of the graph of NRM lost versus partial TRM gained gives the ratio of ancient field to laboratory field. The Thellier experiment was used in the present study mainly as a test of the TRM nature (or otherwise) of the NRM, but the palaeointensities are of considerable interest for their own sakes.

In the initial trial run of eight specimens, relatively rapid double heatings to temperatures of 150, 250, 350, 400, 450, 490, 520, 540, 555 and 570°C were used. Higher temperatures were of little interest since thermal demagnetization had shown that magnetite with blocking temperatures $\leqslant 570$°C was the major NRM carrier in all samples. In the second and third runs (eight specimens each), four more heating steps were added above 400°C and longer individual heatings were used to ensure uniform and identical specimen temperatures in each pair of double heatings. Unfortunately chemical alteration increased as a result of the greater length and number of heatings, and the results were less successful than in the trial run.

AF and thermal demagnetization data and NRM demagnetization results from Thellier experiments were displayed and analysed on both stereographic and orthogonal vector projections (Zijderveld 1967). The latter method is particularly useful because it preserves information about both the direction and the intensity of NRM. Results were considered acceptable if NRM vectors of individual specimens reached a stable endpoint grouping at higher AFs or temperatures, with $<20°$ divergence between the directions recorded by different specimens of the same sample, or if NRM vectors swung in a coherent fashion upon demagnetization. In the latter case, the path followed was usually a great circle, the trace of a bivectorial resultant. Great circles were fitted by eye to individual NRM swings and their point of intersection (Halls 1976; Bailey & Halls 1984) gave an estimate of one of the two vectors (usually the more stable one).

Baked contact test results

The results of the contact test were discouraging. The NRMs of virtually all the marble samples were at or below the noise levels of the spinners. The only exceptions were baked samples from site M2 at the northern contact with the Glanmire marble. Two examples are illustrated in Fig. 3. The results are noisy because of the weak moments of these specimens, but the NRM vector of GL6/2 unmistakably swings from an initial orientation near the present Earth's field (PEF) towards the NW and up '*A*' direction that is the characteristic endpoint of most Tudor gabbro samples. GL5/2 has a very soft NRM that is initially NW and shallow, like typical Grenville '*B*' directions, and swings to negative inclinations at 10 and 15 mT. Even if we were able to demonstrate convincingly that all marbles from the baked zone are overprinted by the Tudor *A* remanence, there is no indication of a stable pre-intrusion remanence in samples outside the baked zone.

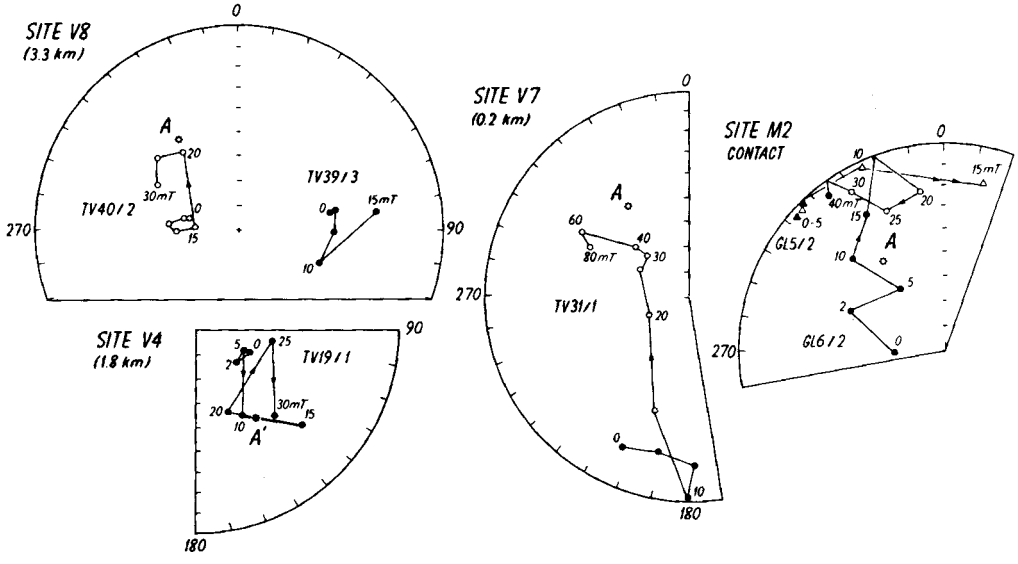


Figure 3. Examples of stable and swinging NRM vectors in country-rock samples that resemble the *A* or *A'* NRMs of the Tudor gabbro. Distances quoted are measured from the intrusive contact. NRMs are very weak and noisy in all marble samples. NRMs are stronger and more stable in volcanic samples, but *A*-type remanences occur erratically both within and beyond the baked contact zone.

Results from the traverse in the Tudor volcanics (sites V1–V9) were more easily measurable, but the contact test was again inconclusive. There were numerous stable NRM vectors but no particular consistency in their directions, except for a small population of Tudor-gabbro-like *A* directions or the reverse of these directions (*A'*). The latter are suspect because there are few normal-polarity *A'* results from the Tudor gabbro itself.

A further problem is that *A* or *A'* directions occur haphazardly in the volcanics, near the baked contact (e.g. site V7), just outside the metamorphic aureole (e.g. site V4), and far beyond the baked zone (e.g. site V8, 3.3 km north of the contact). One possibility is that the gabbro and the volcanics they cut bear a common magnetic overprint, the gabbros being better recorders than the volcanics. The *A* remanence would then date from the time of regional metamorphism and not from the time the Tudor gabbro was intruded. We do not favour this interpretation because the *A* remanence in stably magnetized Tudor gabbro samples has the properties of a pre-metamorphic TRM, as we shall show later.

We propose instead that contact overprinting in and even beyond the mapped contact metamorphic zone was thermochemical rather than purely thermal. This is a common situation near intrusive contacts (Dunlop & Schutts 1983; K. L. Buchan 1984, private communication). We note that Tudor gabbro samples from sites 3, 4 and 8 at or very near the contact are severely altered and gave uninterpretable palaeomagnetic results, whereas samples from all but one of the interior sites gave stable, consistent results. Circulation of hydrothermal solutions along the intrusive contact during emplacement therefore seems probable.

On the basis of the contact test results alone, it is not possible to choose between (i) contact overprinting of NRM in some, but not all, contact rocks at the time the Tudor gabbro was emplaced and acquired a primary *A* TRM, and (ii) later regional overprinting of both gabbro and country rocks. The results reported in the rest of this paper favour the first alternative.

Stable NRM directions in the Tudor gabbro

The NRMs of 42 Tudor gabbro samples from eight sites approached a stable endpoint direction after the elimination of low coercivity or low blocking temperature magnetizations. The sample means (each one the average of three or four specimen endpoint vectors) and site means for this characteristic *A* NRM vector are plotted in Fig. 4. Statistical data appear in Table 1. The palaeopole, 133°E 12°N, is similar to the palaeopole, 137°E 17°N, reported by Palmer & Carmichael (1973). The precision is improved, presumably because cleaning methods available at present remove secondary magnetizations more efficiently. In the present study, $k = 132$ and $\alpha_{95} = 4.8°$, compared to $\alpha_{95} = 8.4°$ in Palmer and Carmichael's work.

In the course of demagnetization, the NRM vectors of 60 Tudor gabbro specimens followed great circle paths, without necessarily reaching a stable *A* endpoint. Fig. 5(a)

Table 1. Results of a three-tier statistical analysis of AF and thermally cleaned *A* magnetization directions. Three or four specimen endpoint magnetizations were averaged to obtain each sample mean (not listed); in turn, sample means were averaged to obtain the site means listed and site means were averaged to give the grand mean with its corresponding palaeopole. *N* is the number of samples or sites averaged; *D*, *I* are declination and inclination of magnetization; *k* is the Fisher precision; and α_{95} is the radius of the 95 per cent confidence cone about the mean direction.

Site	*N*	*D*(°)	*I*(°)	*k*	α_{95}(°)
1	4	334	−52	69	11.0
2	4	329	−49.5	55	13.5
6	6	314	−45.5	77	7.5
7	4	323.5	−42.5	110	8.5
9	6	330.5	−47.5	22	14.5
10	6	329.5	−41	67	8.0
11	6	317.5	−37	31	12.0
12	5	331	−49	43	14.0
Mean, unit weight to sites	**8**	**326**	**−46**	**132**	**4.8**
Palaeopole		133E	12N	($dp = 4°, dm = 6°$)	
Palmer & Carmichael (1973)					
Palaeopole	10	137E	17N	($\alpha_{95} = 8.4°$)	

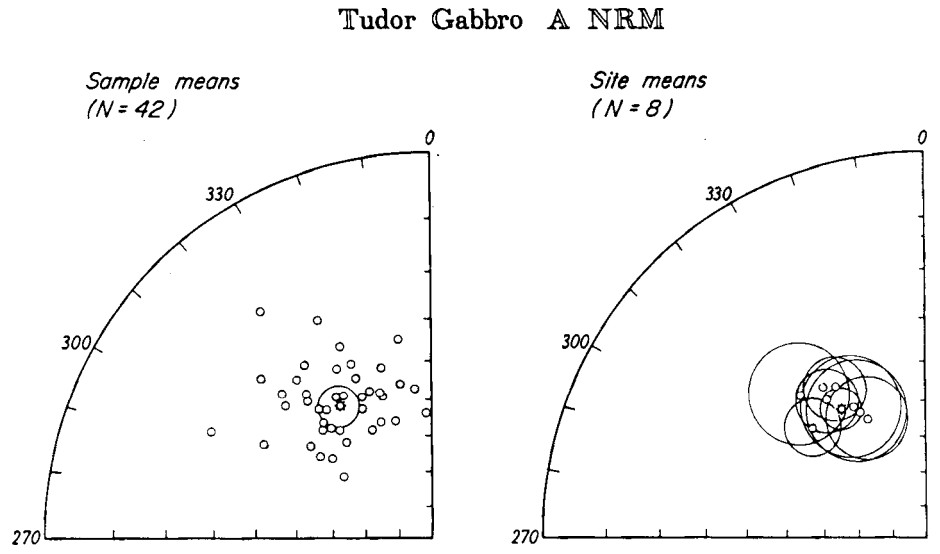


Figure 4. Stereographic projections of sample-mean and site-mean endpoint *A* magnetizations isolated in Tudor gabbro samples. Circles of 95 per cent confidence are shown for site means and grand means (stars).

Table 2. Analysis of remagnetization circles intersecting near the *A* direction of Table 1. The parameter k' corresponds to the Fisher k of Table 1 and the confidence oval semi-axes β_{95} and γ_{95} are analogous to α_{95}. The best intersection point is found by the method of Halls (1976) with statistical analysis following Bailey & Halls (1984).

	N	$D(^\circ)$	$I(^\circ)$	k'	$\beta_{95}(^\circ)$	$\gamma_{95}(^\circ)$
A, unit weight to specimen normals	60	322	−49	29	4.5	6.0
Palaeopole		136E	8N	($dp < 5.5^\circ$, $dm < 8^\circ$)		

illustrates a selection of these remagnetization circles (Halls 1976), which are traces on the unit sphere of the plane defined by the hard (*A*) and soft NRM vectors. The normals to all remagnetization planes are given in Fig. 5(b), together with the intersection point of the great circles (representing the *A* vector, which is common to all remagnetization planes) with its oval of confidence. Statistical parameters estimating the precision with which the intersection point or vector has been located (Bailey & Halls 1984) appear in Table 2. The estimates of the *A* vector from stable endpoint magnetizations and from intersecting remagnetization planes are indistinguishable (Fig. 5b, Tables 1 and 2).

The good definition of the intersection point in Fig. 5 is due in part to the widely dispersed azimuths of remagnetization circles. A small number of circles (about 10) cross the horizontal plane at westerly to southerly azimuths and seem to record superimposed normal and reverse polarity *A* vectors. A larger number (about 15) describe short swings from a north to north-westerly shallow direction, resembling the *B* vector of previous Grenville studies, to the *A*. The remainder (about 35) cross the horizontal with generally northerly azimuths and have not been noted in previous Grenville work.

A detailed view of a few of these latter demagnetization trajectories appears in Fig. 6. We initially thought that the second vector in these bivectorial resultants might be the *C*

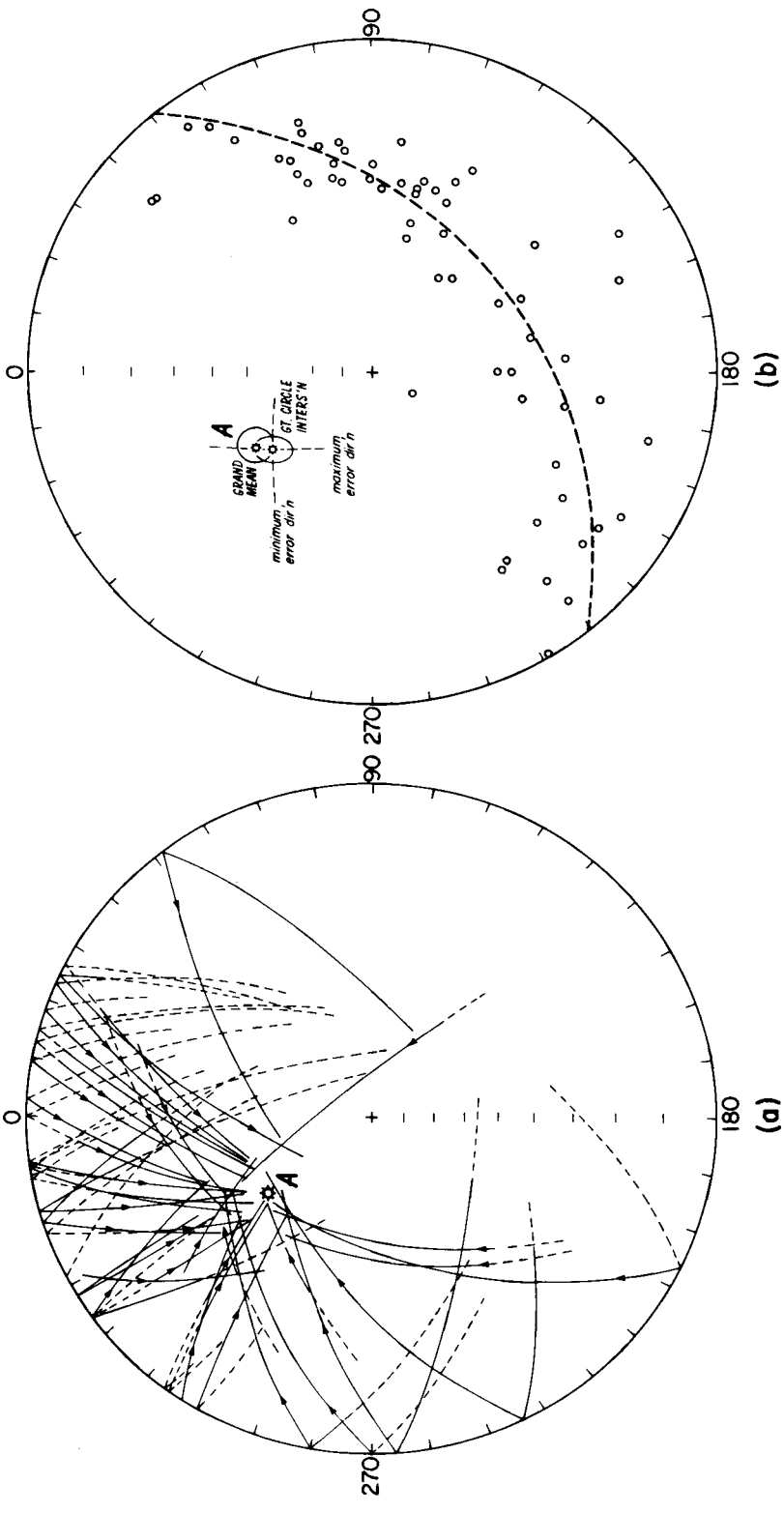


Figure 5. An alternative determination of the Tudor gabbro *A* direction (star), using the convergence of remagnetization circles like those shown in the stereographic projection in (a). Solid segments of great circles are defined by measured step-demagnetization data (for typical examples, see Figs 7–10), while dashed segments join low-coercivity or low T_B subtracted vectors to measured data. Upward normals to all specimen remagnetization planes are indicated by open circles in (b). The normal to the best-fitting great circle through all specimen normals gives the great-circle intersection point of (a). It agrees well with the grand mean of endpoint vectors (Fig. 4).

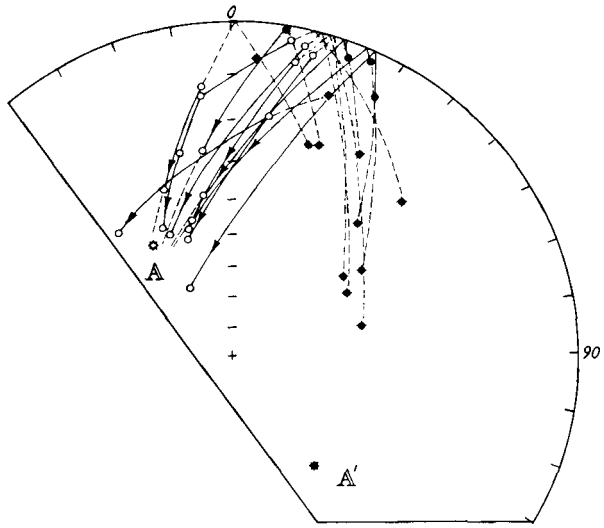


Figure 6. An enlarged view of some of the great circles of Fig. 5(a) that cross the horizontal plane with northerly declinations. Initial and final measured magnetization vectors are shown as circles (open or closed for upper or lower hemisphere respectively) joined by solid great-circle segments. Dashed segments are back projections of these great circles to the directions of low coercivity or low T_B subtracted vectors (diamonds) determined from orthogonal vector diagrams as in Figs 7–10. A' is the downward or normal vector corresponding to A. Subtracted (or 'C') vectors are streaked along a great circle joining A and A'.

magnetization discovered by Dunlop & Stirling (1985) in numerous samples of the Cordova gabbro. Many of the Tudor gabbro NRM vectors, before demagnetization, have declinations just east of north and shallow, usually negative inclinations; the Cordova C vector is usually southerly with shallow positive inclination, although a certain number have the opposite polarity.

Upon demagnetization, this idea was shown to be quite wrong. The bivectorial nature of the NRM was clearly revealed by orthogonal vector diagrams (shown in the next section) and the softer vectors turned out to have directions streaked along the great circle joining A' and C. These results illustrate the danger inherent in trying to recognize component magnetizations by methods that use vector resultants (e.g. point density contouring of resultants, Stupavsky & Symons 1982), rather than resolved component vectors.

Proof of thermal overprinting: blocking temperature spectra

The A magnetization of Tudor gabbro samples is carried mainly by magnetite with discrete high blocking temperatures (T_Bs). TG6/C, 49/2/1 and 60/1/2 (Figs 7, 8 and 9) are typical examples. In these samples and 27 others like them, the orthogonal vector diagram of thermal demagnetization data is ideal: both vector components define linear segments that intersect sharply between 515 and 525°C.

Such perfectly non-overlapping T_B spectra of soft and hard components are seldom observed. They imply that the softer component is a thermal overprint of the A magnetization, and that remagnetization took place in nature at a maximum temperature T_R that corresponds to a laboratory T_B, during rapid heating and cooling, of 515–525°C. Referring to the magnetite blocking contours of Pullaiah *et al.* (1975) or Dunlop (1981a), a reasonable estimate for T_R is 470°C if the subsequent cooling was extremely slow (a few °C Ma^{-1}) and

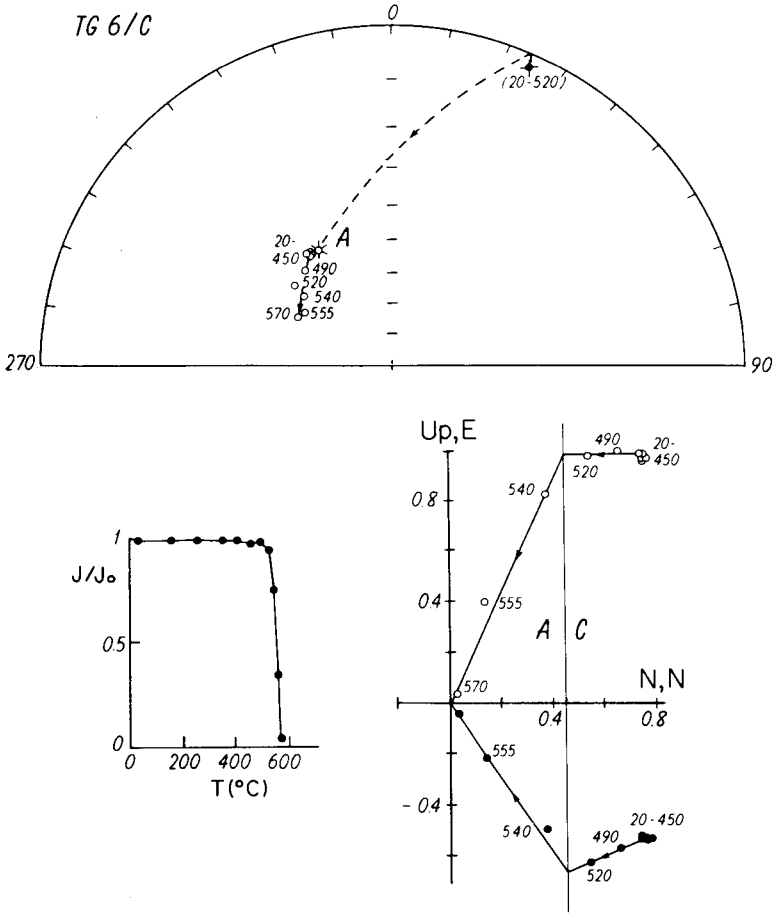


Figure 7. Stereoplot, NRM intensity decay curve, and orthogonal vector projections of stepwise thermal demagnetization data for TG6/C. Here, and in Figs 8–10, upwards and downwards vectors in stereoplots are indicated by open and closed symbols respectively, as they were in Fig. 6. The star is the grand mean *A* direction from Fig. 4. A symbol with a cross denotes a subtracted vector over the temperature range given in parentheses. In orthogonal projections (solid symbols, horizontal or E versus N plane; open symbols, vertical or Up versus N plane; magnetizations in A m^{-1}), it is obvious that a single '*C*' vector is being erased over this temperature range. In all cases, the upper blocking temperature of the soft '*C*' magnetization is 515–525°C and there is no overlap with the high T_B *A* magnetization.

correspondingly higher for faster cooling rates. Since 470°C is about the predicted peak temperature reached by the Tudor body during regional metamorphism, it is logical to suppose that the low T_B remanence is a thermal overprint dating from this time. The higher T_B *A* remanence thus predates regional metamorphism.

The relative intensities and directions of the *A* and overprint vectors vary considerably. TG6/C (Fig. 7) has a soft vector orthogonal to, and about one-third as intense as, the *A* NRM. (The orthogonality of *A* and '*C*' accounts for the lack of any decay in the total NRM of this specimen below 500°C.) In TG49/2/1 (Fig. 8), the *A* and overprint vectors are about equally intense and are >90° apart. The overprint vector in TG60/1/2 (Fig. 9) is twice as intense as the *A* and their directions are <90° apart. But in all cases, the *A* vector has T_Bs

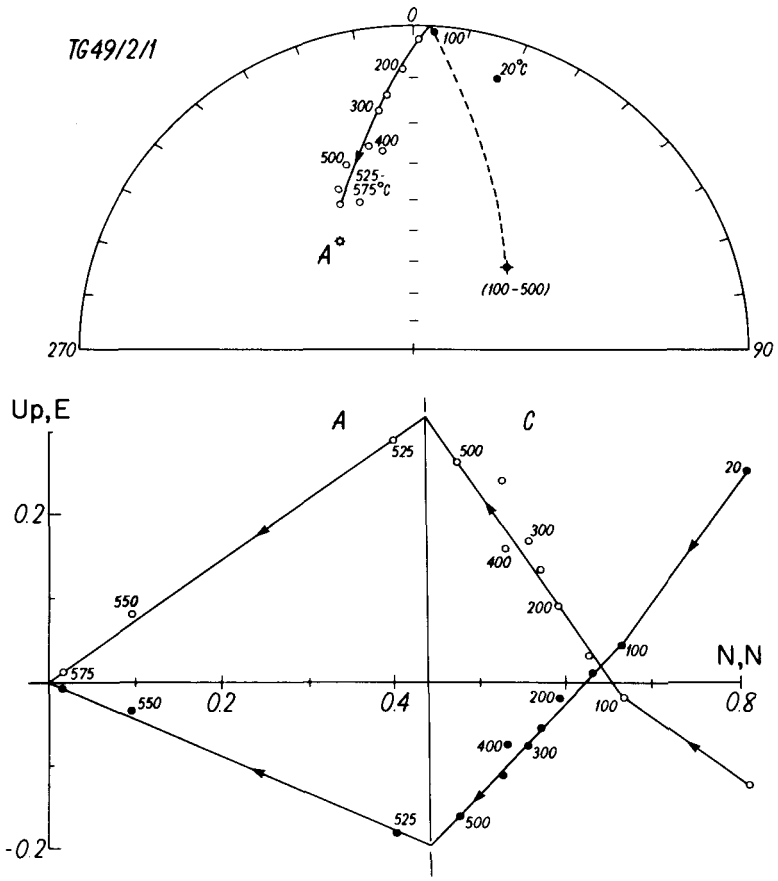


Figure 8. Another example of ideally partitioned blocking-temperature spectra of A and C magnetizations, in TG49/2/1. Here the C magnetization has a higher intensity and a broader T_B spectrum than in Fig. 7, and the temperature separating the A and C T_B spectra is 515°C.

exclusively above 520°C or so, while the overprinting 'C' vector has T_Bs entirely below about 520°C.

The situation is rather different when A and A' vectors are superimposed. The A' NRM definitely has lower T_Bs on average than the A, but since the vectors are very nearly reversed to each other, quite complicated overlaps of T_B spectra cannot be ruled out. In the case of TG1/A1 (Fig. 10), A' is carried principally by magnetite with T_Bs below 550°C but A resides in both magnetite and haematite with T_Bs as high as 640°C. The A' remanence seems to be associated with samples in which haematite is a significant NRM carrier, and it is possible that partial self-reversal of primary TRM is recorded rather than a field reversal during cooling.

Proof of primary TRM: Thellier palaeointensity determination

Thellier-type palaeointensity determinations were attempted for fresh specimens of 24 samples, as described earlier. In most cases, the A remanence formed 85 per cent or more of the total NRM in these specimens, although a few determinations were attempted on rocks

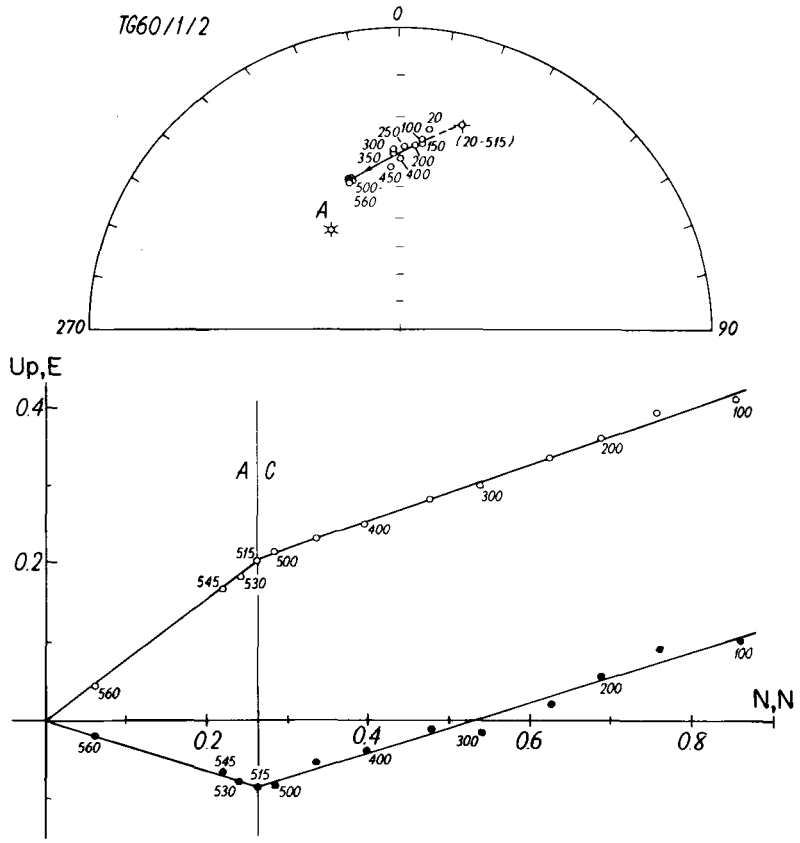


Figure 9. A third example of non-overlapping T_B spectra of *A* and *C* magnetizations, with a common temperature of 515°C, in TG60/1/2. In this specimen, the *C* magnetization is larger than the *A* and has a very broad T_B spectrum.

with bivectorial NRMs. As a preliminary experiment, both NRM and laboratory-produced weak-field (60 μT) anhysteretic remanent magnetization (ARM) were AF demagnetized for fresh companion specimens of some samples and their intensity decay curves were compared (Fig. 11). The purpose was to see if the coercivity spectrum of ARM, which is analogous to that of TRM, matched the coercivity spectrum of the NRM. This was indeed the case for most specimens tested and lends some support to the idea that the NRM is a TRM and therefore primary.

Better evidence comes from the palaeointensity runs themselves. A selection of results appears in Fig. 12. TG2/B2, 4/B1, 9/C1 and 52/3 have essentially univectorial NRMs, TG6/C and 60/1/3 (cf. Figs 7 and 9) have bivectorial NRMs, and TG32/3 has a trivectorial NRM in which a present field (PEF) overprint is removed up to about 350°C. In all cases, it is again clear that the *A* NRM has no blocking temperatures below about 520°C. This fact presents a practical problem in the palaeointensity experiment, in that the entire *A* remanence is demagnetized over a 40–50°C range. Because of the difficulty of reproducing temperatures exactly in double heatings, temperature steps of less than 15°C were not feasible. Linearity must therefore be judged on the basis of only four data points, the highest temperature one often being perturbed by sample alteration (e.g. TG60/1/3). Longer equilibration times were

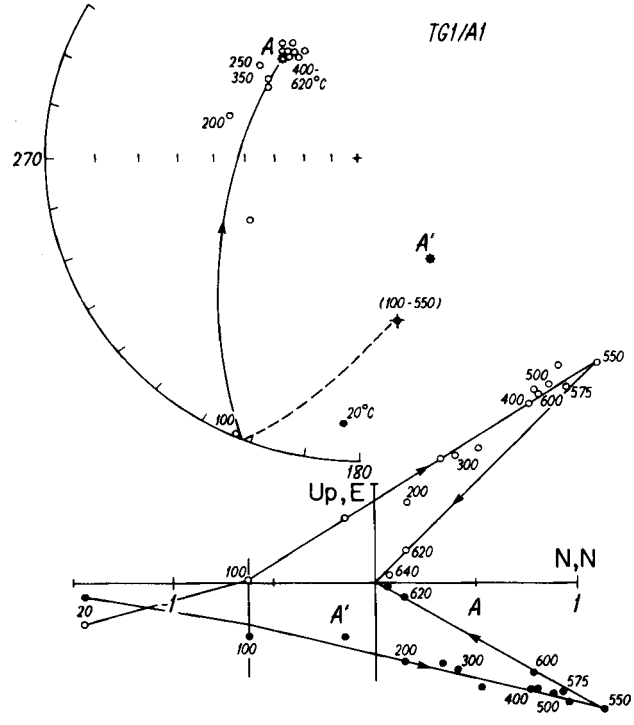


Figure 10. An example of superimposed, nearly antipodal A and A' magnetizations in TG1/A1. A' and A have apparently non-overlapping T_B ranges of 100–550°C and 550–640°C respectively. However, undetected overlap of T_B spectra is possible with vectors like A' and A that are nearly coaxial. Note that haematite is an important carrier of the A remanence in this specimen; its role was negligible in samples with B and C magnetizations.

used in the later runs to improve temperature uniformity and reproducibility, but as pointed out earlier, chemical alteration was then accentuated.

Given these limitations, the better data (mainly from the first run) define acceptable straight lines. TG2/B2, 4/B1, 6/C and 9/C1 are examples of such data. The palaeointensity estimates given by the slopes of these graphs appear in Table 3. They are in the range 18 ± 3 μT, with three of the values clustering near the mean. These intensities are about one-half of the present field intensity at a magnetic latitude of 27°, which is the palaeolatitude of the Tudor gabbro at the time the A remanence was acquired.

Data from the second and third runs are more scattered (e.g. TG32/3, 52/3, 60/1/3) and lines through them are anchored rather arbitrarily by the first and last data points. These results would certainly not be taken as *prima facie* determinations of palaeofield intensity by any reasonable set of criteria, but it is interesting that the values listed in this second group in Table 3 are reasonably consistent with values from the first, more trustworthy group. Five of the six are between 17 and 27 μT. TG60/1/3, which altered severely above 545°C, gives the only decidedly anomalous value: 35 μT.

In cases where the C component was erased over four or five successive temperature steps, a palaeofield determination was made by comparing the intensity decay of the C vector with the pTRM gained over the same range. The procedure is not as straightforward as it is for the characteristic or highest T_B component; recourse must be made to orthogonal vector plots

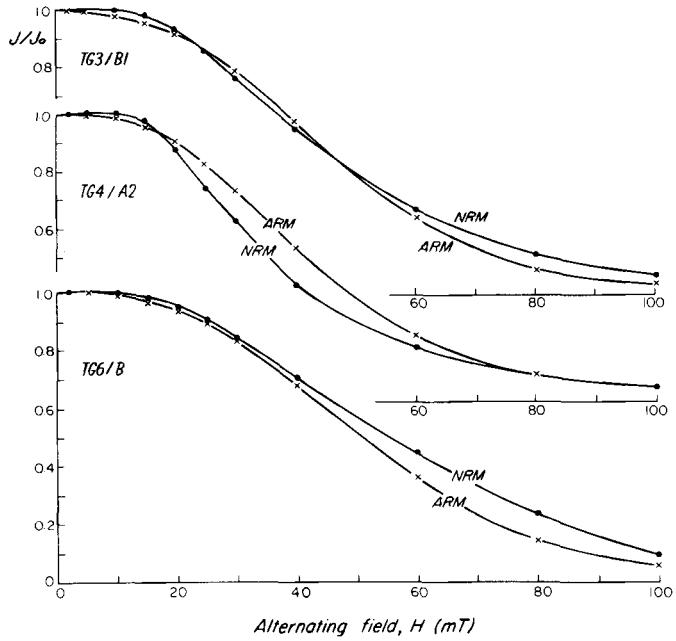


Figure 11. Comparison of intensity decay curves of ARM (60 μT steady field) and univectorial NRM for several pilot specimens used in Thellier palaeointensity runs. The similar, hard NRM and ARM decay curves indicate that NRM is carried by a single mineral phase (single domain size magnetite) and was probably acquired as a TRM.

like those of Figs 7 and 9. The *C* component palaeointensities were scattered and will not be considered further.

We view the Thellier palaeointensity results on the *A* component of NRM as a qualified success. In those samples that did not alter significantly, the TRM nature of the *A* NRM is fairly certain, and in the remaining samples, some degree of linearity between NRM and pTRM was preserved until alteration became pronounced.

Palaeopoles and Grenvillian continental drift

Despite the failure of the baked contact test, it seems fairly certain that the characteristic *A* magnetization of the Tudor gabbro is the high blocking temperature portion of a primary TRM. It has survived later thermal overprinting during Grenvillian regional metamorphism and must predate the metamorphism. Since hornblendes from the Tudor closed to Ar diffusion at a temperature of about 590°C around 1110 Ma (Baksi 1982), and the magnetic blocking temperatures range from 570 to about 520°C (Figs 7–9, 12), a maximum age for the *A* remanence is about 1100 Ma. A minimum age of about 1050 Ma is imposed by the oldest known ages, about 1000 Ma, of post-Grenvillian uplift and cooling magnetizations (Fig. 13). The age of Grenvillian metamorphism, in the Tudor area at least, is bracketed by the same dates.

How then is one to interpret the Tudor gabbro palaeopole (Fig. 13), which must now be regarded as older, not younger, than other Grenvillian poles? Poles from Interior Laurentia (mainly from the Lake Superior Keweenawan succession) in the age interval 1100–1050 Ma

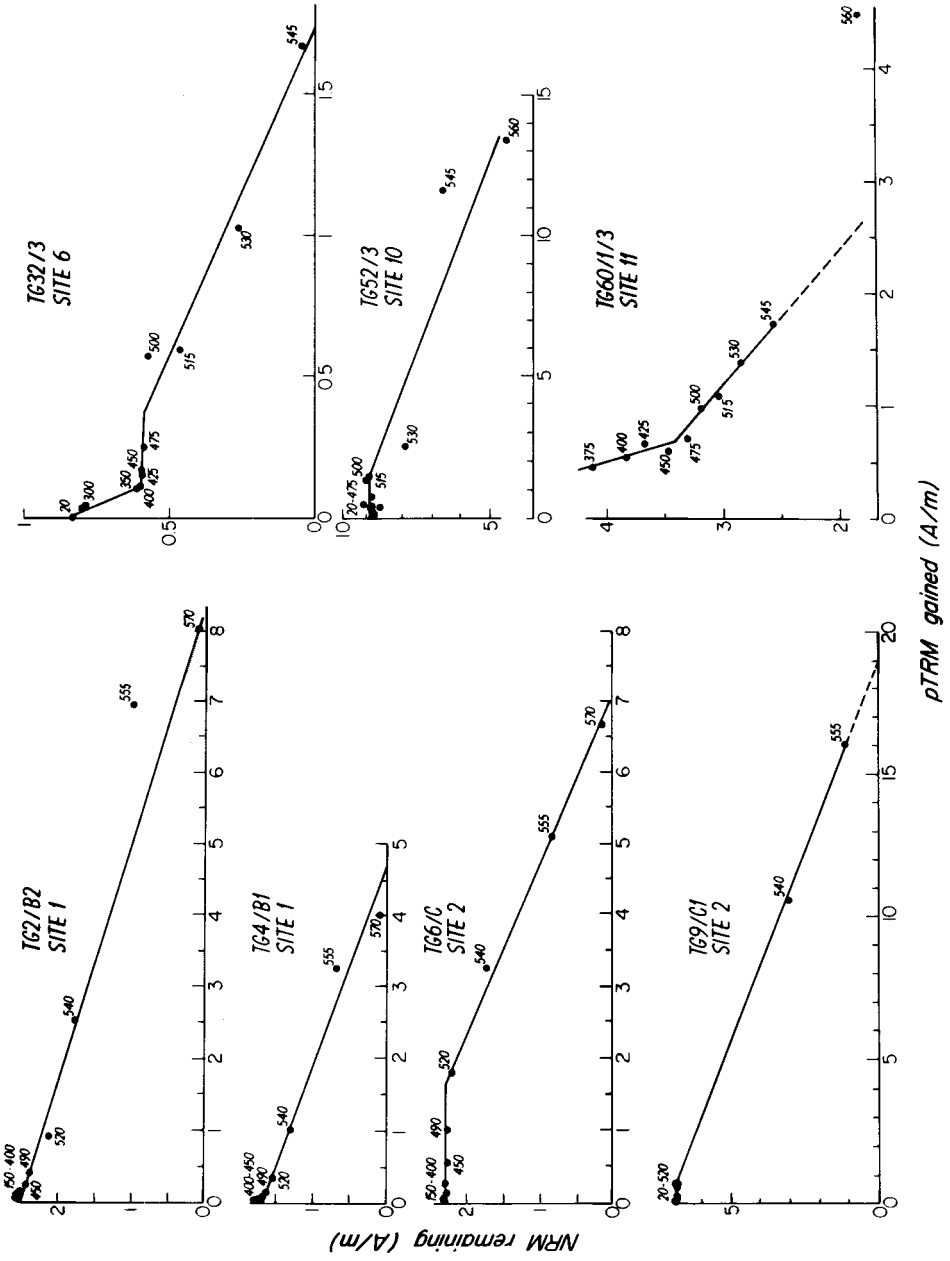

TG2/B2
SITE 1
TG4/B1
SITE 1
TG6/C
SITE 2
TG9/C1
SITE 2
TG32/3
SITE 6
TG52/3
SITE 10
TG60/1/3
SITE 11
NRM remaining (A/m)
pTRM gained (A/m)

Table 3. Estimates of palaeofield intensity H_P determined from the A magnetization by the Thellier double-heating method. Results in the first group are more reliable than those in the second (cf. Fig. 12).

Specimen	T_B range (°C)	H_P (μT)
TG 2/B2	490–570	15
3/C	490–555	17.5
4/B1	490–570	18
6/C	520–570	21
9/C1	490–555	18.5
32/3	500–545	21.5
33/3	500–545	24
52/3	500–560	18
55/2/1	500–545	17
60/1/3	475–545	35
63/3/1	500–560	27

trend along the 180° meridian from 30°N to the equator. Poles Tu and Tu_A are in obvious disagreement with these well-established poles.

One tectonic model that would reconcile the results is shown in Fig. 14. In this reconstruction, the Tudor A magnetization was impressed while a small ocean (<1000 km) existed between ancestral Grenvillia and Interior Laurentia. Because of the indeterminacy of palaeolongitude, a host of similar reconstructions are possible, but all require a pole of closing near the (present-day) eastern tip of Grenvillia. That is, the discrepancy between the Grenvillian and other Laurentian data can be largely accommodated by a small-angle rotation of Grenvillia with respect to Interior Laurentia, rather similar to that suggested for the north-eastern Grenville by Roy & Fahrig (1973) on quite different grounds. (A local rotation of the Tudor gabbro and its surrounding country rocks would of course also accommodate the results, but there is no geological evidence that this occurred.)

The cartoon of Fig. 14 would imply that the collisional phase of the Grenvillian orogeny followed emplacement of the Tudor gabbro and acquisition of the A magnetization during initial cooling. This picture is inconsistent with the fact that the Tudor volcanics and Hastings and Glanmire sediments (Fig. 2) were already steeply dipping at the time they were intruded by the Tudor gabbro. A more reasonable model is that collision first occurred at or before 1200 Ma, that the Tudor gabbro and related stocks were intruded after deformation was complete, but that convergence continued for another 100 Myr or so with extensive underplating of the Interior Laurentian protocontinent by Grenvillian crust, in the manner

Figure 12. Examples of Thellier-type palaeointensity results. On the left are the relatively reliable results of the initial run, in which heatings and coolings were comparatively rapid and samples did not alter appreciably below 570°C. The apparent lack of NRM decay in TG6/C between 150 and 520°C, a range in which a sizeable pTRM is gained, is a result of the approximate orthogonality of A and C magnetizations; as Fig. 7 demonstrates, although the total NRM intensity is almost constant, the C magnetization is substantial and decays completely over this temperature range. On the right are less reliable results from univectorial (TG52/3), bivectorial (TG60/1/3) and trivectorial (TG32/3) specimens, most of which underwent significant chemical alteration at and above 545°C as a result of longer heatings. Despite the greater scatter of points, average palaeointensity values for most of the less reliable specimens are similar to the more reliable determinations (see Table 3).

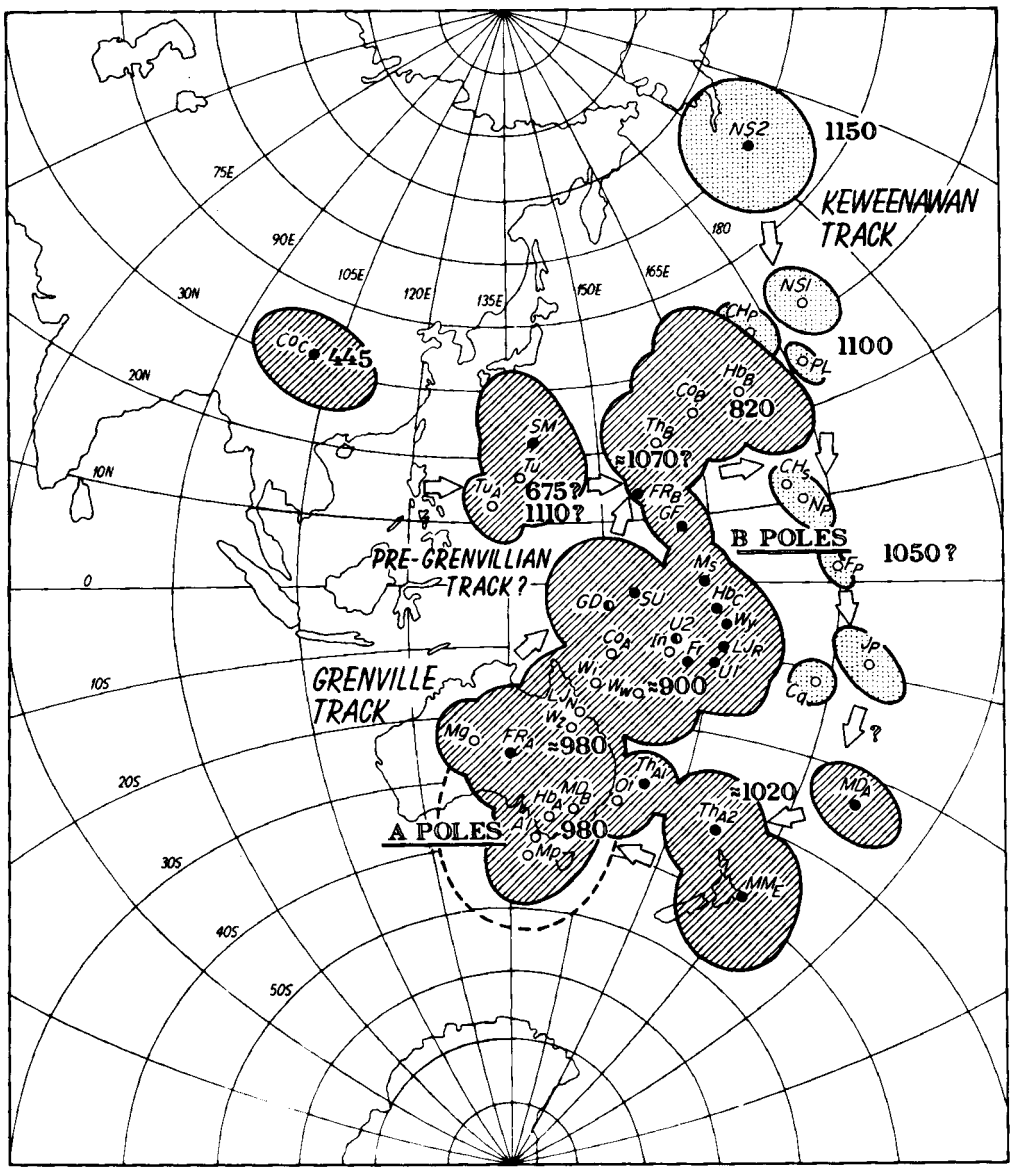


Figure 13. Palaeopoles for the Tudor gabbro (Tu: Palmer & Carmichael 1973; Tu_A: *A* remanence, this study) compared with Grenville *A* and *B* poles dating from post-metamorphic uplift and cooling between 1020 and 820 Ma and with selected 1100–1050 Ma Keweenawan poles from Interior Laurentia. (A complete listing of all poles illustrated is given in Dunlop & Stirling 1985.) The Tudor poles disagree with both Grenvillian and Interior Laurentian poles. For the reasons given in the text, Tu_A is believed to be a pre-metamorphic remanence about 1100 Myr old. It therefore defines a short pre-Grevillian metamorphism track for Grenvillia that converges with the Interior Laurentian APWP around 1050 Ma, which is thus the best estimate for the time at which convergence between Grenvillia and Interior Laurentia ceased and post-metamorphic uplift began.

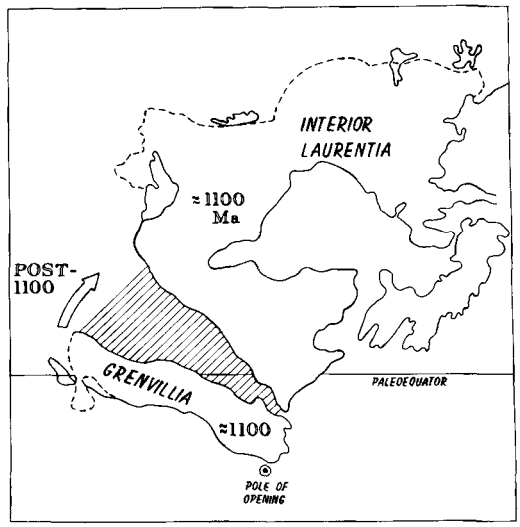


Figure 14. One possible reconstruction of the relative positions of Grenvillia (as preserved at present) and Interior Laurentia around 1100 Ma when the Tudor *A* remanence was acquired. All such reconstructions require subsequent clockwise rotation of Grenvillia about the pole shown, consuming or underplating the hatched region. For convenience in drawing, the Grenville Front is depicted as the northern margin of Grenvillia, but the actual sature with Interior Laurentia is known to lie at least 100 km to the south.

of the more recent convergence between India and the Asian continent. In this picture, the 'ocean' between Grenvillia and Interior Laurentia in Fig. 14 would actually have been Grenvillian crust, later (after 1100 Ma) consumed under Laurentia. The cessation of convergence and the onset of erosion, uplift and cooling must have occurred around 1050 Ma.

The exact chronology of Grenvillian orogenic events is only as certain as the $^{40}Ar/^{39}Ar$ and other ages that pinpoint key events, most notably in the present model the initial cooling of the Tudor gabbro (Baksi 1982). Once the time framework is established, provided pre-burial magnetizations can be demonstrated to have survived, palaeomagnetism places quite close constraints on the relative rotations and palaeolatitude shifts of the colliding continents.

In fairness, we should point out that incorporating the Tudor gabbro pole on a younger extension of the Grenville Loop, as has been done in the past, is a perfectly reasonable join-the-poles approach. Pole SM of the Steel Mountain anorthosite of western Newfoundland (Murthy & Rao 1976), which lies in granulite terrain and has a K/Ar chlorite age of 450 ± 35 Ma, is probably a late post-metamorphic overprint that belongs on such an extension. However, the Laurentian APWP appears to have made two or more loops through this general area between 700 and 400 Ma (Irving 1979; Dunlop 1981b) and we do not regard the agreement of SM with Tu and Tu_A (Fig. 13) as compelling evidence of contemporaneity.

If, despite the contrast in metamorphic conditions, the SM and Tu poles are actually of similar ages, the Tudor *A* magnetization would have to be a low-temperature CRM, since the Tudor was not reheated sufficiently around 1000 Ma, let alone around 700 or 450 Ma, to acquire TRM in the T_B range 520–570°C. The very sharp and consistent upper T_Bs of the overprinting remanences, which we find suggestive of the leading edge of an ≈500°C thermal event, would then be no more than a curious coincidence. We do not favour this interpretation, but the magnetic evidence is not yet compelling enough to rule it out.

Conclusions

(1) The palaeopole of the Tudor gabbro characteristic *A* magnetization, first determined by Hood (1958) and Palmer & Carmichael (1973), is indeed anomalous compared to other Grenvillian poles. Our determination, using stable endpoint magnetizations and intersecting great circles, places the pole at 12°N, 133°E ($dp = 4°$, $dm = 6°$).

(2) The baked contact test with surrounding volcanic and sedimentary rocks was inconclusive. It neither proved nor disproved that the *A* is the primary NRM of the Tudor gabbro.

(3) Thellier-type palaeointensity determinations were successful in samples that did not alter severely during heating and indicated a palaeofield strength about one-half of that of the present-day field at the same latitude. Although this evidence is not conclusive, the *A* magnetization is likely a TRM.

(4) The *A* remanence is overprinted by a secondary magnetization which superficially resembles the *C* magnetization of the Cordova gabbro (Dunlop & Stirling 1985) but is rather erratic in intensity and direction. The upper blocking temperatures of this '*C*' magnetization are consistently between 515 and 525°C and have no overlap with blocking temperatures of the *A* magnetization. The *A* magnetization has therefore been *thermally* overprinted at peak temperatures in nature of 470–500°C, presumably during regional burial metamorphism which locally produced maximum temperatures in this range. The *A* magnetization is thus pre-metamorphic, in contrast to other reported Grenvillian remanences, which date from post-metamorphic uplift and cooling.

(5) The anomalous Tudor gabbro palaeopole records either local rotation or, more likely, pre-metamorphic continent-wide rotation of Grenvillia with respect to Interior Laurentia on a scale sufficient to close a small ocean and underplate continental crust in Himalayan fashion. A tentative chronology of events is as follows: initial collision and deformational phase of orogeny around 1200 Ma; intrusion of post-kinematic plutons like the Tudor before 1100 Ma; continued continental convergence until about 1050 Ma; and slow uplift and cooling lasting until at least 800 Ma.

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References

Baer, A. J., 1981. Two orogenies in the Grenville Belt?, *Nature*, **290**, 129–131.

Bailey, R. C. & Halls, H. C., 1984. Estimate of confidence in palaeomagnetic directions derived from mixed remagnetization circle and direct observational data, *J. Geophys.*, **54**, 174–182.

Baksi, A. K., 1982. ^{40}Ar-^{39}Ar incremental heating studies on the Tudor gabbro, Grenville Province, Ontario: its bearing on the North American apparent polar wander path in late Proterozoic times, *Geophys. J. R. astr. Soc.*, **70**, 545–562.

Berger, G. W. & York, D., 1981. $^{40}Ar/^{39}Ar$ dating of the Thanet gabbro, Ontario: looking through the Grenvillian metamorphic veil and implications for palaeomagnetism, *Can. J. Earth Sci.*, **18**, 266–273.

Berger, G. W., York, D. & Dunlop, D. J., 1979. Calibration of Grenvillian palaeopoles by $^{40}Ar/^{39}Ar$ dating, *Nature*, **277**, 46–48.

Buchan, K. L., 1978. Magnetic overprinting in the Thanet gabbro complex, *Can. J. Earth Sci.*, **15**, 1407–1421.

Buchan, K. L. & Dunlop, D. J., 1976. Palaeomagnetism of the Haliburton intrusions: superimposed magnetizations, metamorphism and tectonics in the late Precambrian, *J. geophys. Res.*, **81**, 2951–2967.

Buchan, K. L., Fahrig, W. F., Freda, G. N. & Frith, R. A., 1983. Palaeomagnetism of the Lac St-Jean anorthosite and related rocks, Grenville Province, Quebec, *Can. J. Earth Sci.*, **20**, 246–258.

Dallmeyer, R. D. & Sutter, J. F., 1980. Acquisitional chronology of remanent magnetization along the 'Grenville polar path': evidence from $^{40}Ar/^{39}Ar$ ages of hornblende and biotite from the Whitestone diorite, Ontario, *J. geophys. Res.*, **85**, 3177–3186.

Dunlop, D. J., 1981a. The rock magnetism of fine particles, *Phys. Earth planet. Int.*, **26**, 1–26.

Dunlop, D. J., 1981b. Palaeomagnetic evidence for Proterozoic continental development, *Phil. Trans. R. Soc. A*, **301**, 265–277.

Dunlop, D. J. & Schutts, L. D., 1983. Proterozoic magnetic overprinting and tectonics of the Laurentian Shield, *Mem. geol. Soc. Am.*, **161**, 35–45.

Dunlop, D. J. & Stirling, J. M., 1985. Post-tectonic magnetizations from the Cordova gabbro, Ontario and Palaeozoic reactivation in the Grenville Province, *Geophys. J. R. astr. Soc.*, **82**, 521–550.

Dunlop, D. J., York, D., Berger, G. W., Buchan, K. L. & Stirling, J. M., 1980. The Grenville Province: a palaeomagnetic case-study in Precambrian continental drift, *Spec. Pap. geol. Ass. Can.*, **20**, 487–502.

Halls, H. C., 1976. A least-squares method to find a remanence direction from converging remagnetization circles, *Geophys. J. R. astr. Soc.*, **45**, 297–304.

Hayatsu, A. & Palmer, H. C., 1975. K-Ar isochron study of the Tudor gabbro, Grenville Province, Ontario, *Earth planet. Sci. Lett.*, **25**, 208–212.

Hood, P. J., 1958. Palaeomagnetic studies of some Precambrian rocks in Ontario, *PhD thesis*, University of Toronto,

Irving, E., 1979. Palaeopoles and palaeolatitudes of North America and speculations about displaced terrains, *Can. J. Earth Sci.*, **16**, 669–694.

Irving, E., Emslie, R. F. & Ueno, H., 1974. Upper Proterozoic paleomagnetic poles from Laurentia and the history of the Grenville Province, *J. geophys. Res.*, **79**, 5491–5502.

Lopez-Martinez, M. & York, D., 1983. Further thermochronometric unravelling of the age and paleomagnetic record of the southwest Grenville Province, *Can. J. Earth Sci.*, **20**, 953–960.

Lumbers, S. B., 1964. Preliminary report on the relationship of mineral deposits to intrusive rocks and metamorphism in part of the Grenville Province of southeastern Ontario, *Ont. Dep. Mines, Prelim. Rep. 1964-4*, 33 pp.

Lumbers, S. B., 1967. Geology and mineral deposits of the Bancroft-Madoc area, in *Geology of Parts of Eastern Ontario and Western Quebec*, pp. 13–19, ed. Jenness, S. E., Geological Association of Canada Guidebook.

Lumbers, S. B., 1969. Geology of Limerick and Tudor townships, *Ont. Dep. Mines, Geol. Rep. 67.*

Murthy, G. S. & Rao, K. V., 1976. Paleomagnetism of Steel Mountain and Indian Head anorthosites from western Newfoundland, *Can. J. Earth Sci.*, **13**, 75–83.

Palmer, H. C. & Carmichael, C. M., 1973. Paleomagnetism of some Grenville Province rocks, *Can. J. Earth Sci.*, **10**, 1175–1190.

Palmer, H. C., Hayatsu, A., Waboso, C. E. & Pullan, S., 1979. A paleomagnetic and K-Ar study of the Umfraville gabbro, Ontario, *Can. J. Earth Sci.*, **16**, 459–471.

Park, J. K. & Emslie, R. F., 1983. Paleomagnetic history of the Mealy dykes of Labrador, Canada, *Can. J. Earth Sci.*, **20**, 1818–1833.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.*, **28**, 133–143.

Roy, J. L. & Fahrig, W. F., 1973. The paleomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador: polar wandering and tectonic implications, *Can. J. Earth Sci.*, **10**, 1279–1301.

Roy, J. L. & Robertson, J. B., 1979. An assessment of the St-Urbain and other Grenville paleopoles, *Can. J. Earth Sci.*, **16**, 1857–1865.

Silver, L. T. & Lumbers, S. B., 1966. Geochronologic studies in the Bancroft-Madoc area of the Grenville Province, Ontario, Canada (abstract), *Spec. Pap. geol. Soc. Am.*, **87**, 156.

Stupavsky, M. & Symons, D. T. A., 1982. Isolation of early Paleohelikian remanence in Grenville anorthosites of the French River area, Ontario, *Can. J. Earth Sci.*, **19**, 819–828.

Symons, D. T. A., 1978. Paleomagnetism of the 1180 Ma Umfraville gabbro, Ontario, *Can. J. Earth Sci.*, **15**, 956–962.

Zijderveld, J. D. A., 1967. A. C. demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism*, pp. 254–286, eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, Amsterdam.